COM 5506
Social Network Analysis for Communication

L7 Community Detection

SEM B 2026

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Reminder

- Proposal due today
- We will discuss today

Project Mode

- Consultation 1: today – only talk about data and rough idea
- Between today and Apr 2, get your data
- Consultation 2: show me your data, and tell me how it goes. I will let you know if you data look okay
- Between consultation 2 and consultation 3: work on analysis
- Consultation 3: discuss your analysis and findings.
- Between consultation 3 and final presentation: complete a draft for report, and PowerPoint presentation
- After presentation, further improve your work. And email your final work to your TA. (one week after presentation)

Why do we look for community structure?

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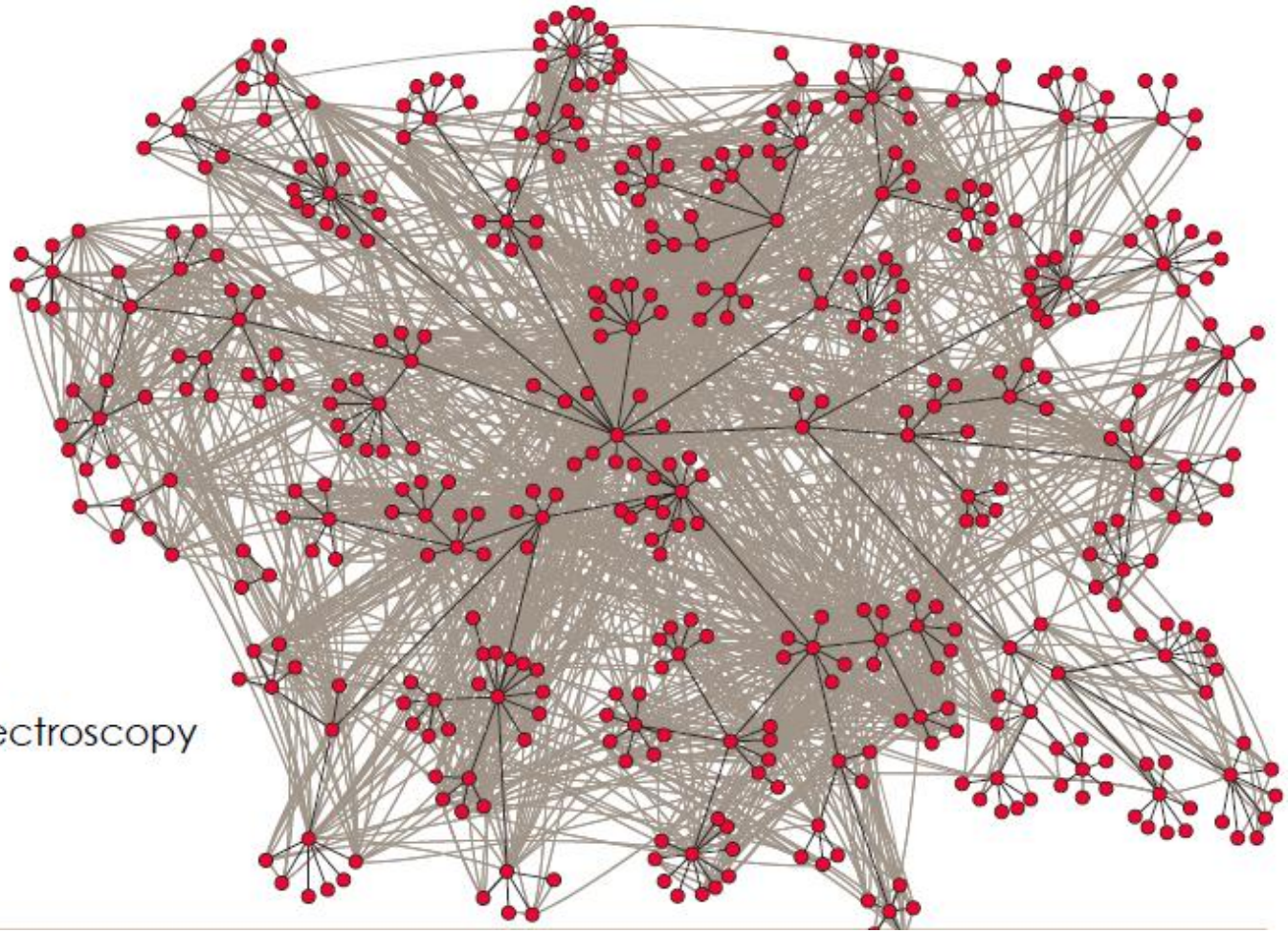
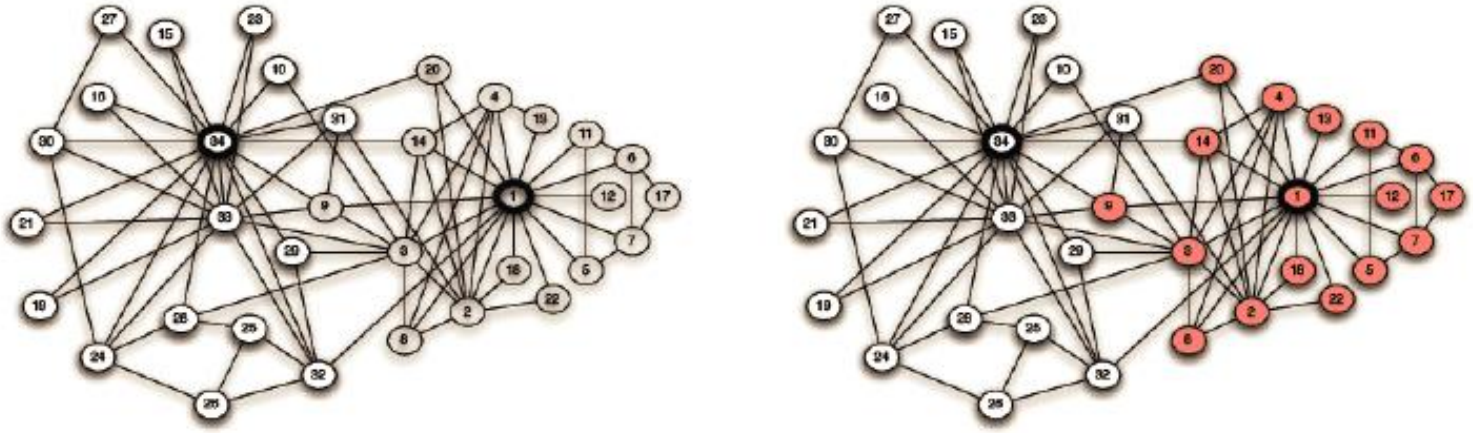


example:
email spectroscopy

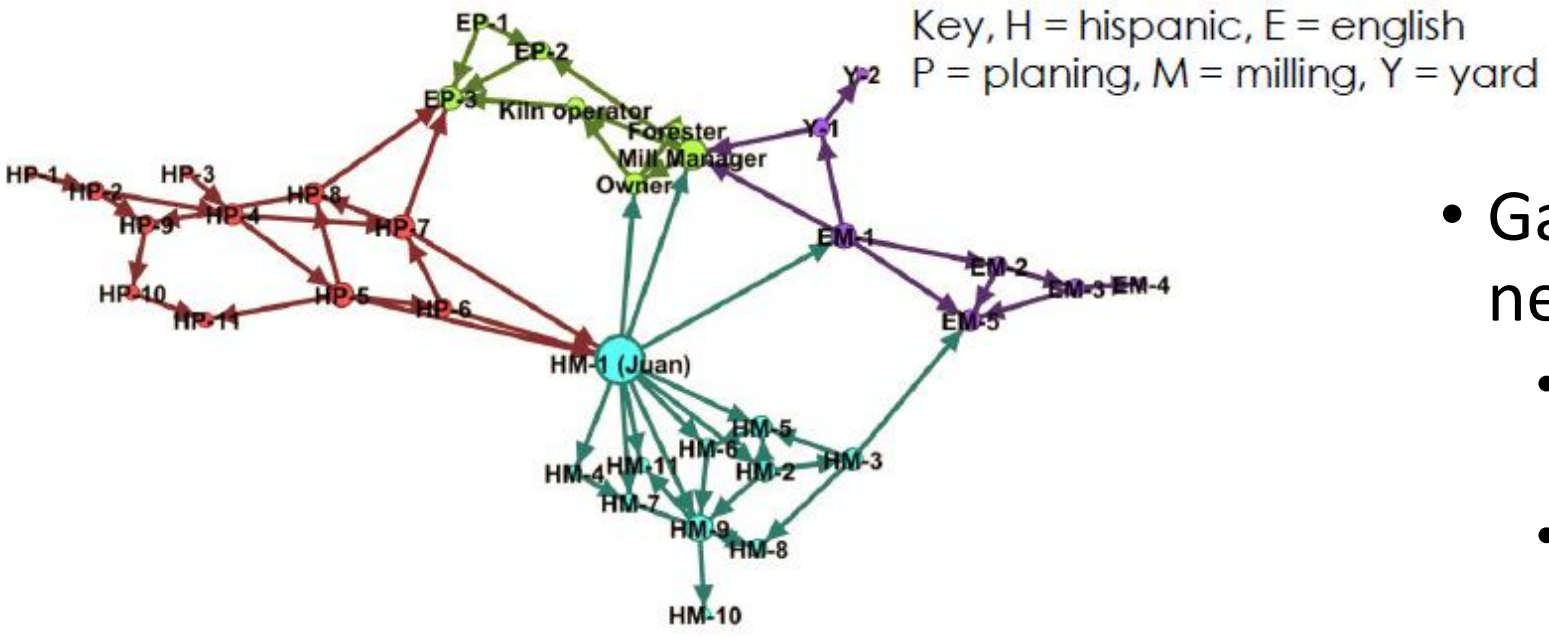
Zachary Karate Club



(a) *Karate club network* (b) *After a split into two clubs*

source:Easley/Kleinberg

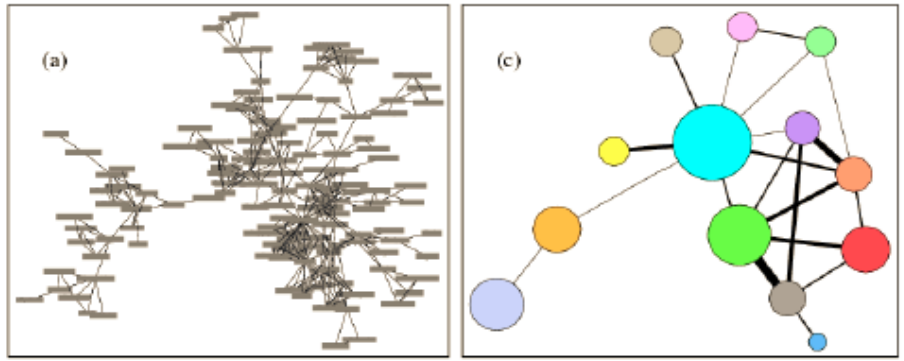
Reason 1 – understanding

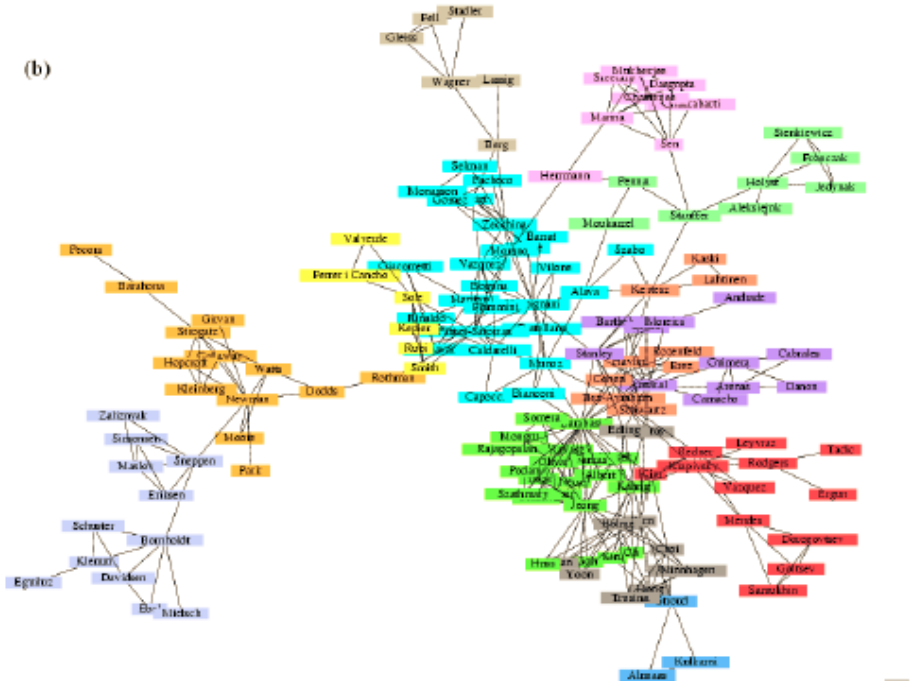


At a sawmill, Hispanic workers are Reluctant to adopt a new plan.

- Gain understanding of networks
  - Discover communities of practice
  - Measure isolation of groups
  - Understand opinion dynamics/adoption

Reason 2 – visualization





- Communities help to ‘aggregate’ network data
- Help to present data more efficiently
- Help to reduce individual data to group data

Example

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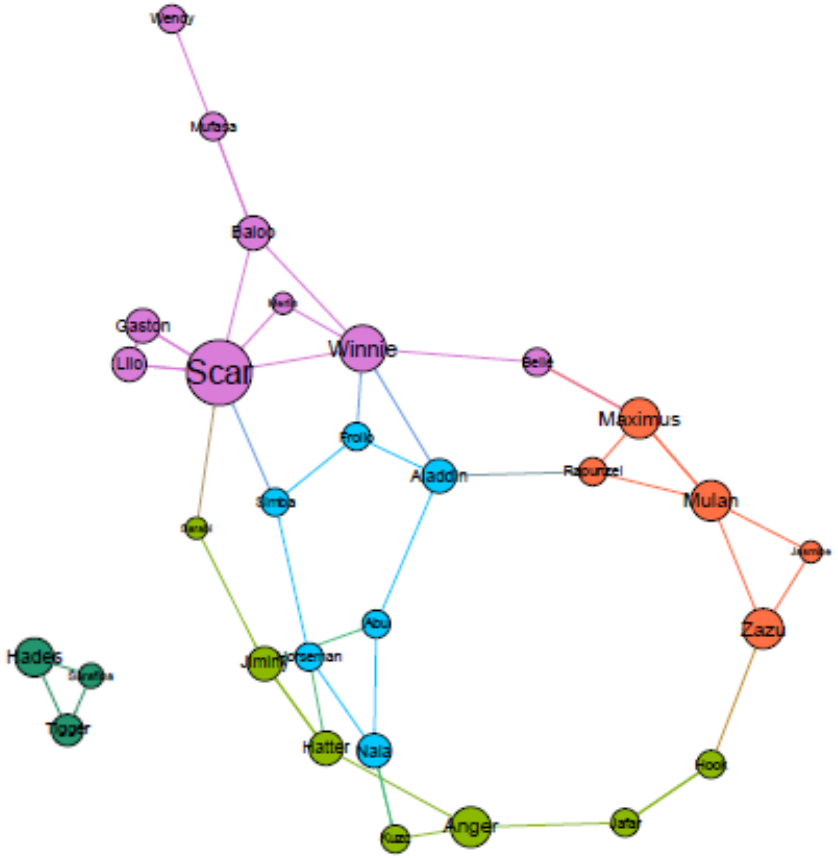

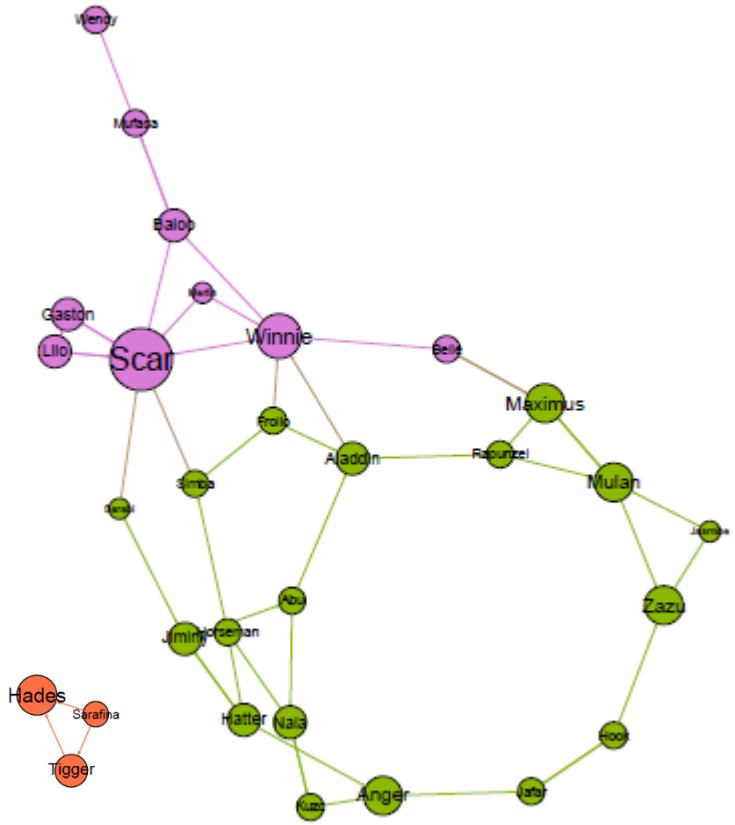

What makes a community?

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- Mutuality of ties (regular fully connected graph)
  - Everyone in the group knows everyone else
- Frequency of ties among members (degree centrality)
  - Everyone in the group has links to at least k others in the group
- Closeness or reachability of subgroup members (small world)
  - Individuals are separated by at most n hops
- Relative frequency of ties among subgroup members compared to nonmembers (discriminant power)

Component vs. Community

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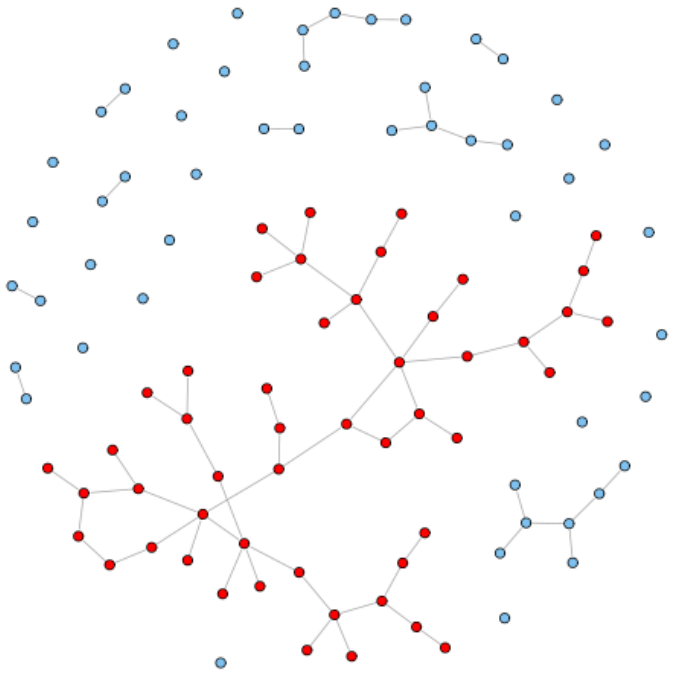
	Component	Community
Strengths	A "hard" boundary (based on observable edges) exists between components	A network can be partitioned into multiple communities for local-level analysis
Weaknesses	Large social networks usually have a giant component (≈ the network)	A "soft" boundary (based on hidden "similarities") is used to define communities

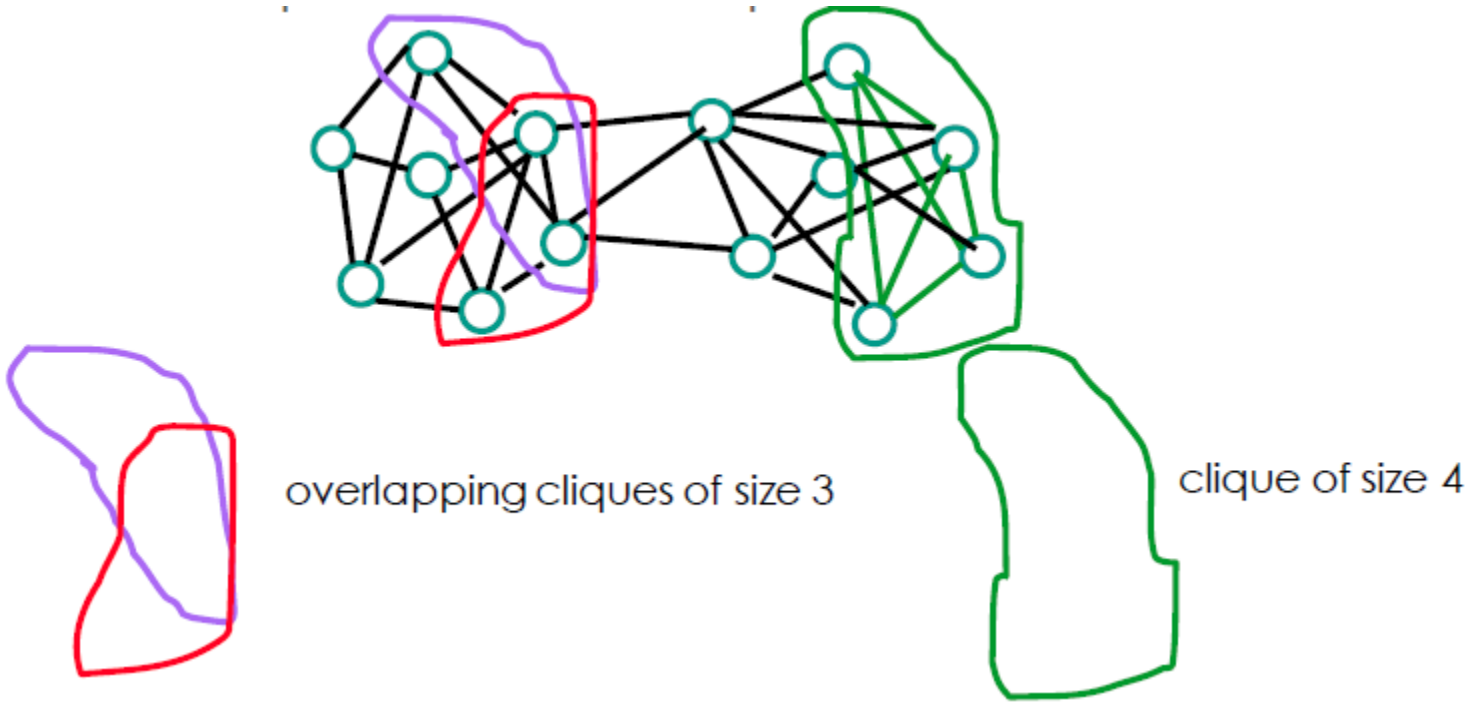
Component not very useful



- Usually giant component is the part we want to focus our analysis on.
- We need to know the structure of the red part.
- We are not very interested in the separation of red and blue parts.

Cliques – a very stringent criterion

- Every member of the group has links to every other member
- Cliques can overlap



But, cliques are not very much informative

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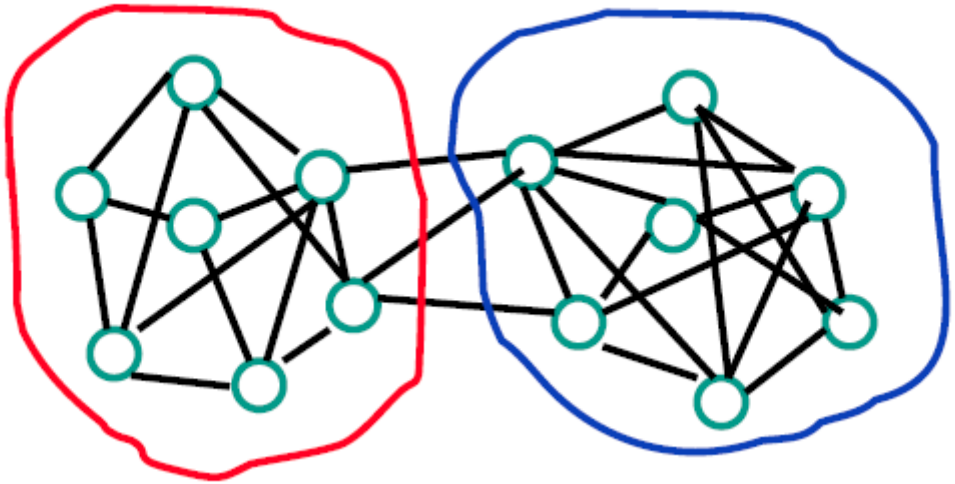
- No robust
  - One missing link can disqualify a clique
- Not interesting
  - Everyone is connected to everyone else
  - No core-periphery structure
  - No centrality measures apply
- How cliques overlap can be more interesting than that they exist

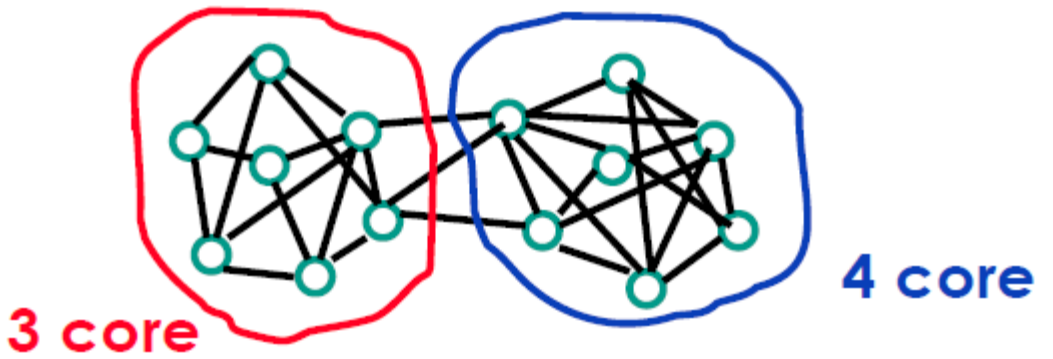
K-cores – a less stringent criterion

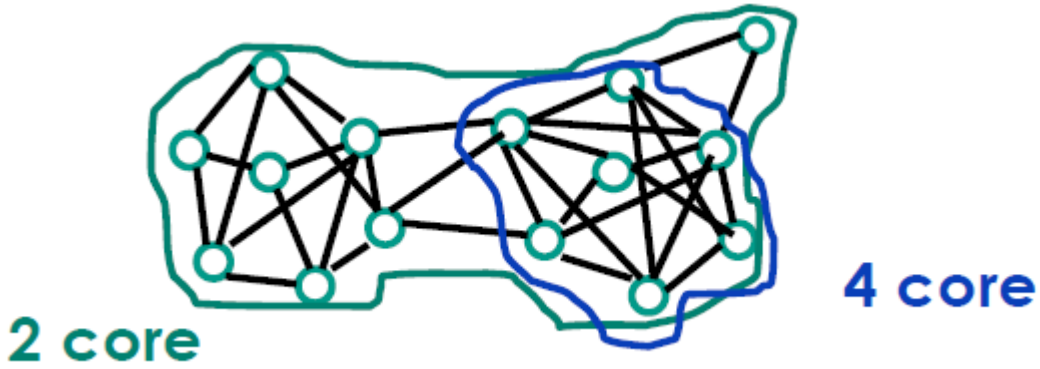
- Each node is connected to k other nodes in the group
- What is the k for the red core?
- What is the k for the blue core?



K-cores

- Still this is a too stringent requirement for identifying natural communities





Community detection requirement

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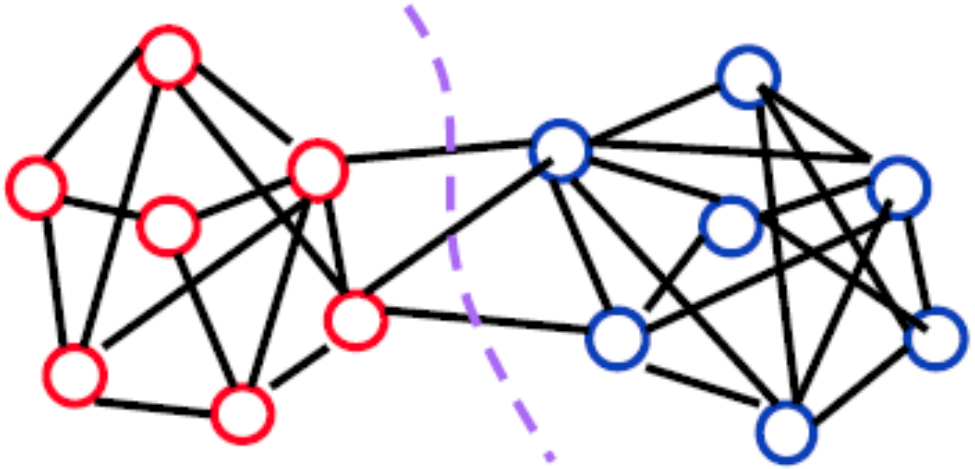
- Social and other networks have a natural community structure
- We want to discover this structure rather than impose a certain size of community or fix the number of communities
- Without looking, can we discover community structure in an automated way?



Cluster analysis

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- A cluster is a collection of nodes that are “similar” to each other.
- Cluster analysis partition the network into K number of groups (clusters) with members being maximally similar within each group and maximally dissimilar between groups
- Cluster analysis is an unsupervised learning (i.e., without predefined group membership) as opposed to supervised classification (with group membership predefine, e.g., logistic regression or discriminant analysis)

Supervised vs. unsupervised learning

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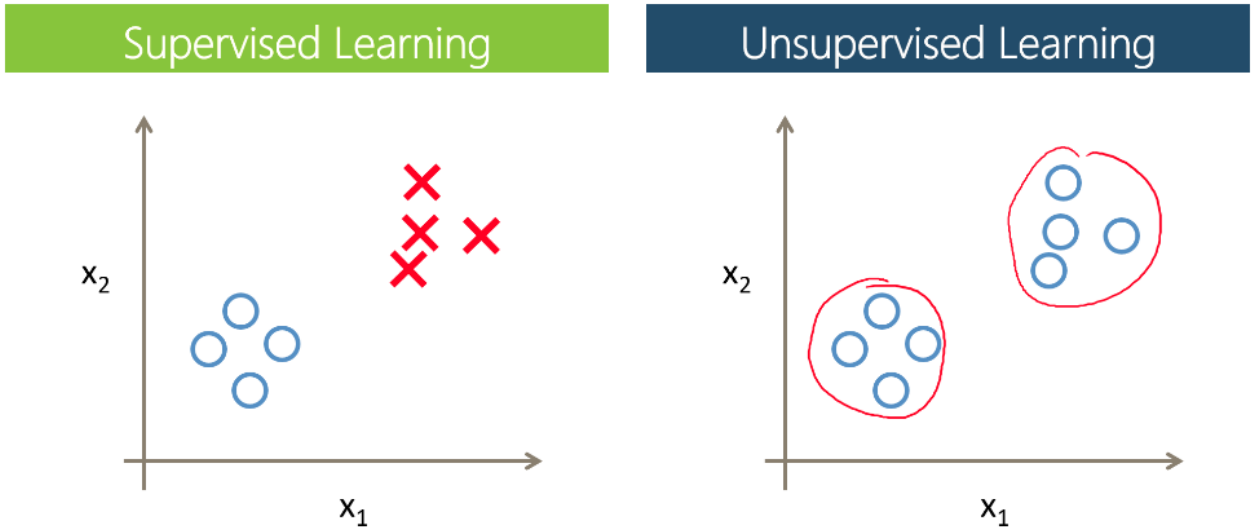
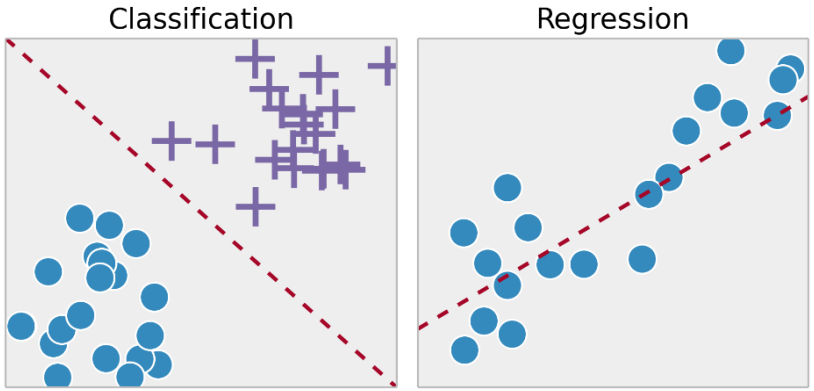
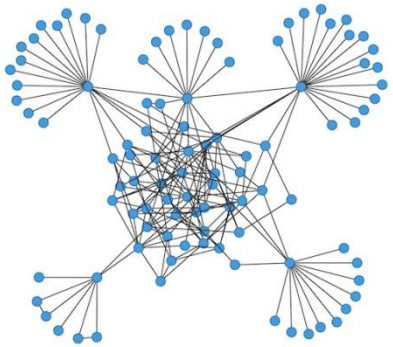
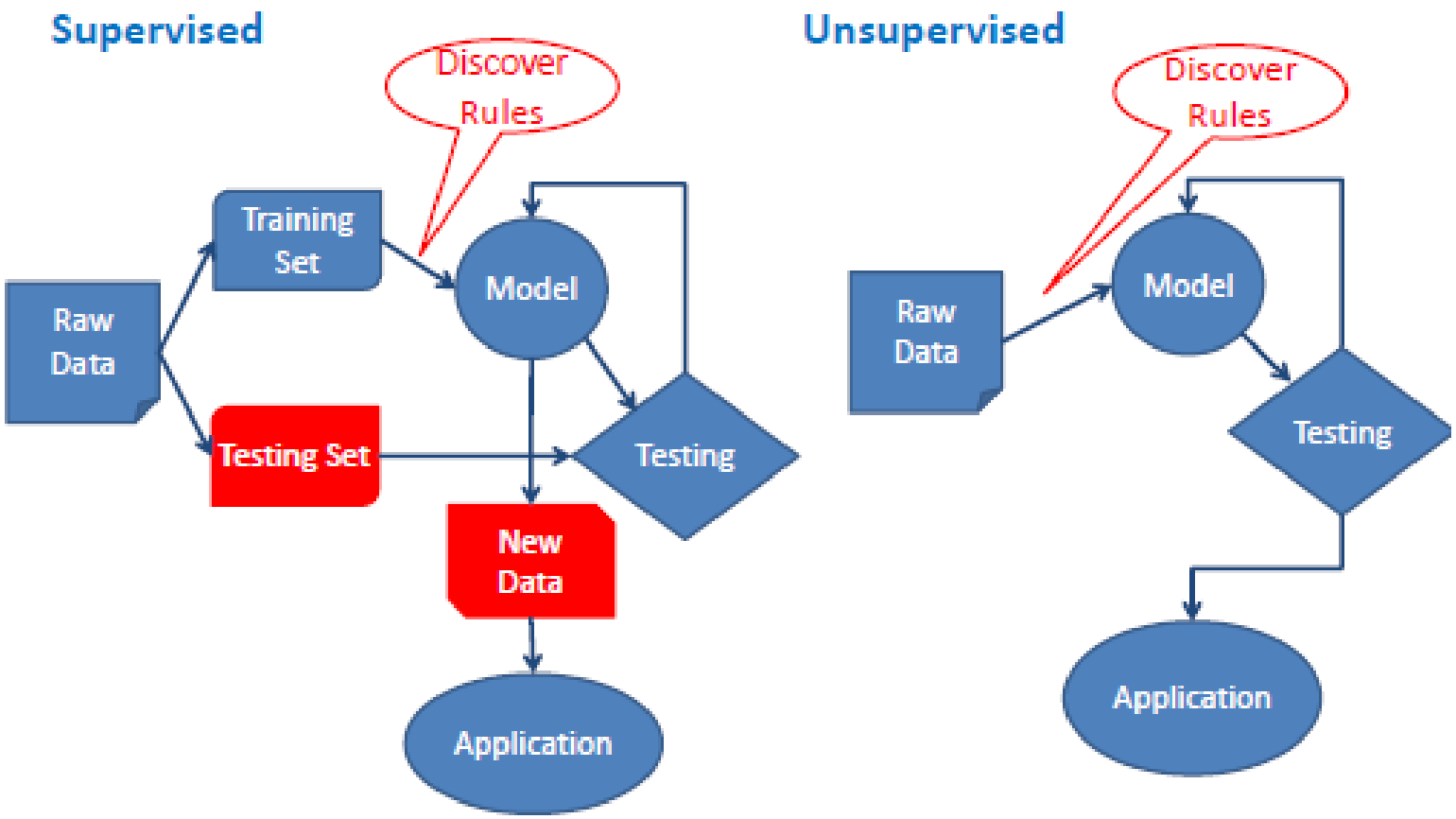


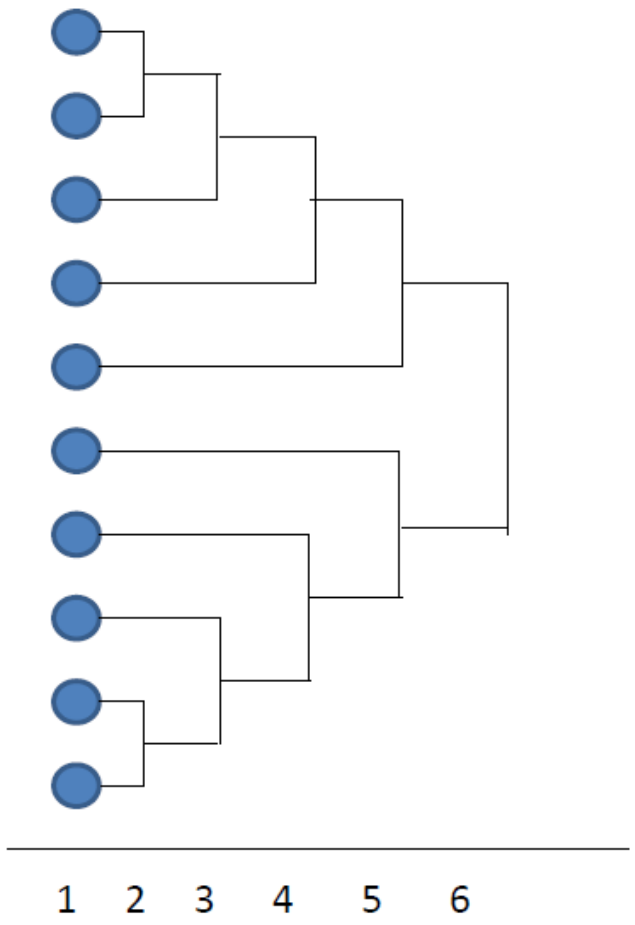




Supervised vs. unsupervised learning



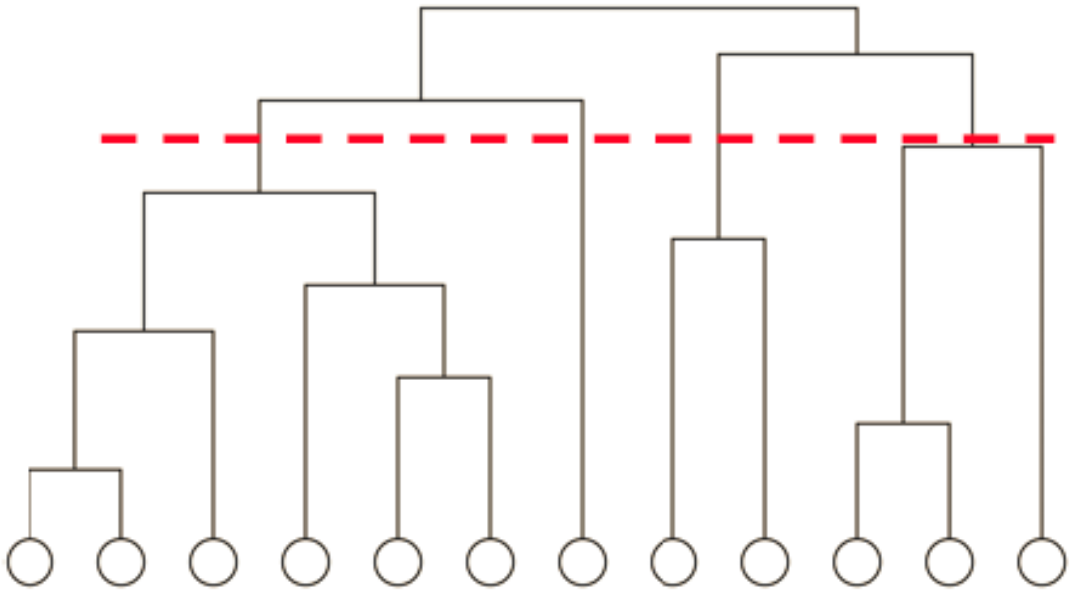
Dendrograph process of cluster analysis



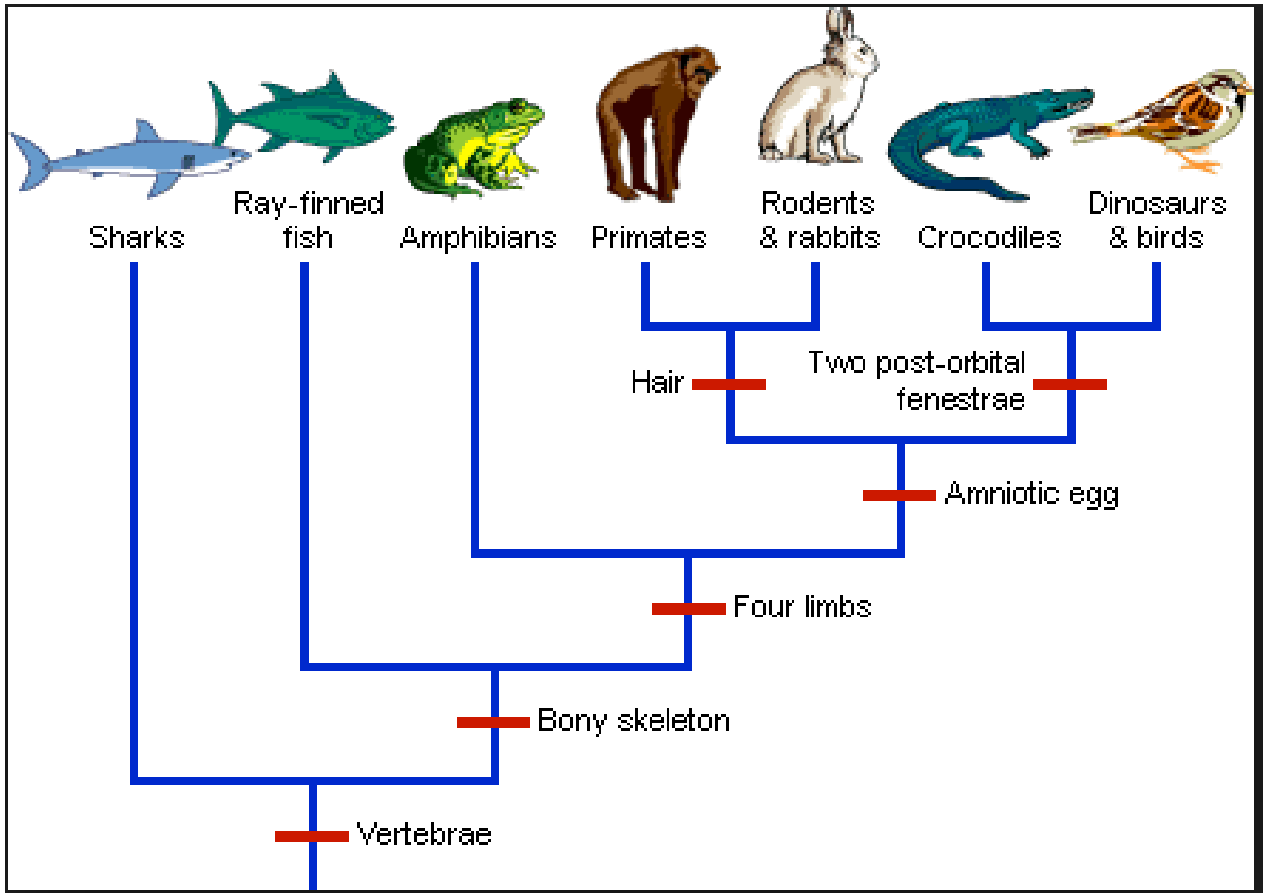
- At step 1, there are 10 clusters, each with a node that is uniquely different from all others.
- At step 2, mode 1 and node 2 are considered to be similar enough to form a cluster; same thing for node 9 and 10; now there are 8 clusters.
- At step 3, node 3 joins the cluster of 1 and 2, and node 8 joins the cluster of 9 and 10.
- The process keeps until every node is included in the giant cluster at step 6.
- An optimal solution is to keep a small number of clusters with maximal similarity within and maximal difference between.

Hierarchical clustering

- Process:
  - After calculating the “distances” for all pairs of nodes
  - Start with all n nodes disconnected
  - Add edges between pairs one by one in order of decreasing weight
  - Nested components, where one can take a ‘slice; at any level of the tree



Cluster Dendrograph



Betweenness clustering

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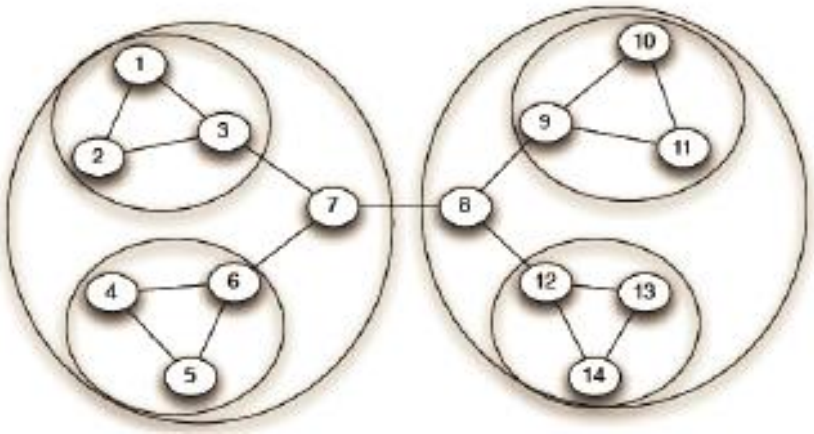
- Algorithm
  - Compute the betweenness of all edges
  - While (betweenness of any edge> threshold):
    - Remove edge with highest betweenness
    - Recalculate betwenness

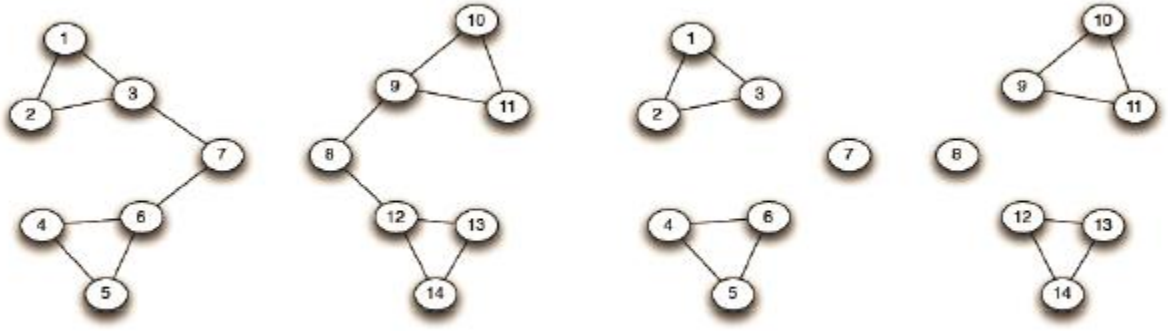


- *Betweenness Centrality (BC)*
- A measure of the "intermediacy" role of node

$$C_B(i) = \sum_{j<k} g_{jk}(i)/g_{jk}$$

where g_{jk} is the number of shortest paths between all pairs of j and k ($j \neq i \neq k$) and $g_{jk}(i)$ is the number of the shortest paths that pass through i.

Replace i with edge x.



(a) *Step 1* (b) *Step 2*

When to partition network?

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- Graph Modularity: Modularity is one measure of the structure of networks or graphs. It describes the appropriateness of breaking the graph into communities, to justify the need for local level analysis

Modularity

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- Modularity is the fraction of the edges that fall within the given groups minus the expected such fraction if edges were distributed at random.
- There are different formulas to calculate modularity.
- A high Q score of modularity indicates nodes are highly connected within each community but sparsely connected between different communities.
- For a random network, Q=0. in other words, the number of edges within a community is no different from what you would expect.
- As a rule of thumb, a modularity larger than 0.3 indicates that the partition being assessed appropriately describes the underlying clustered structure of the network.

Lab 6

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- Introduction to Gephi